

Multispectral satellite imagery for illegal waste material classification

State of the art and thesis direction

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The problem: priority, not just presence

Illegal waste dumping (discariche abusive) is an environmental crime and a public-health issue.

Agencies such as ARPA cannot inspect everything: with limited inspectors, they must prioritise - and priority depends on WHAT is dumped, the material.

Automated pipelines now find sites well (detection is essentially mature) - but they classify presence, not what is there.

The advisor's own survey (Fraternali et al. 2024, 50 works, 1987-2023): nearly all RGB; material identification declared an open gap requiring ≤ 30 cm imagery + multispectral data.

Research question

What is the added value of multispectral data, vs. RGB only, for waste material classification from satellite imagery?

Why material matters: $\text{risk} = \text{hazard} \times \text{exposure} \times \text{magnitude}$

- The hazard is material-bound: inert rubble vs plastics vs asbestos-cement are worlds apart.
- Material carries the hazard: the European Waste Catalogue (EWC) flags hazardous codes with * - asbestos-containing construction material is 17 06 05*.
- Priority risk = hazard (material) x exposure (who/what is nearby) x magnitude (how much).
- Regione Lombardia already operationalises this for asbestos roofs: the Indice di Degrado (d.d.g. 13237/2008) score triggers re-evaluation, 3-year remediation, or 12-month removal.
- **So identifying the MATERIAL - not just locating the site - is the decision-relevant question.**

Today's paradigm: RGB deep learning on aerial imagery

VHR aerial / satellite tile (RGB) → CNN or Transformer backbone → pretraining (ImageNet, EO SSL) → two-step fine-tune → binary waste / no-waste.

- Accurate within the geographic context it was trained on: Gibellini 2025, F1 92.0% at 20 cm.
- Classifies presence, not material.
- Tied to chromatic information only.
- Generalizzazione collapses outside it: -5.1% F1 cross-region; worse cross-sensor.

These limits motivate looking at the spectrum.

State of the art: RGB waste detection

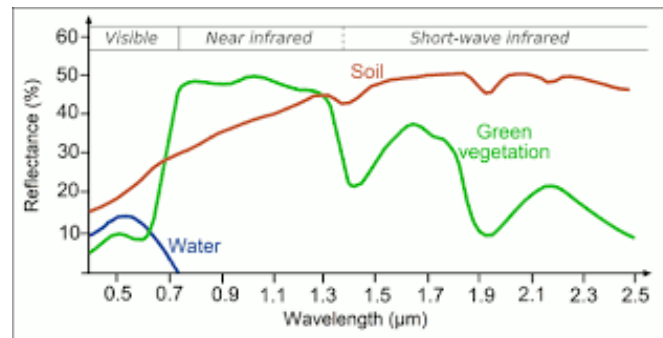
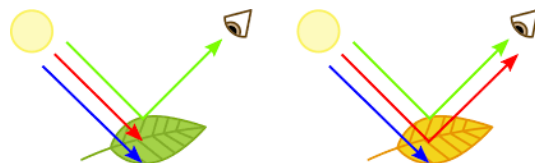
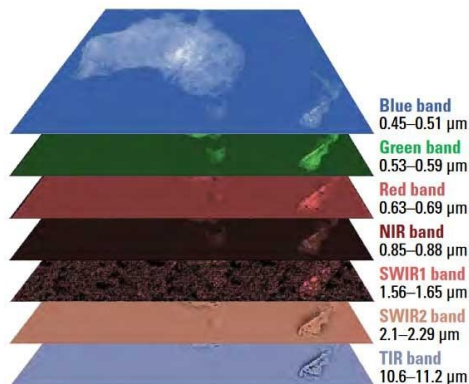
Work (year)	Input / GSD	Task	Method	Key result
Gibellini 2025	AerialWaste RGB 20 cm	classification	Swin-T + RSP, two-step	F1 92.0%; cross-region -5.1%
Disaitek 2024	Pleiades Neo 0.3 m	detection	operational service	waste ≥ 2 m ² ~95% (vendor)
CascadeDumpNet 2024	Pleiades 0.5 m RGB	object detection	cascade CNN + AutoML	mAP 84.6%; transfers across cities
Sun 2023 (global)	VHR RGB 0.3-1 m	object detection	BCA-Net (Faster R-CNN)	~2,500 dumpsites; sens. 98%
AerialWaste (Torres 2023)	aerial RGB 20-50 cm	dataset + cls.	ResNet+FPN baseline	F1 80.7%; 22 categories, RGB-only

A satellite pixel is a spectrum, not just a colour

Each pixel is a vector of reflectance values, the material's spectral signature.

- RGB, 3 broad visible bands (colour only).
- Multispectral (MS), 4-15 chosen bands (e.g. WorldView-3 8 VNIR + 8 SWIR; Pleiades Neo 6 VNIR).
- Hyperspectral (HSI), hundreds of contiguous narrow bands (EnMAP 230; USGS splib07a 2,151 ch).

This vector is the raw input from which any classifier reasons.

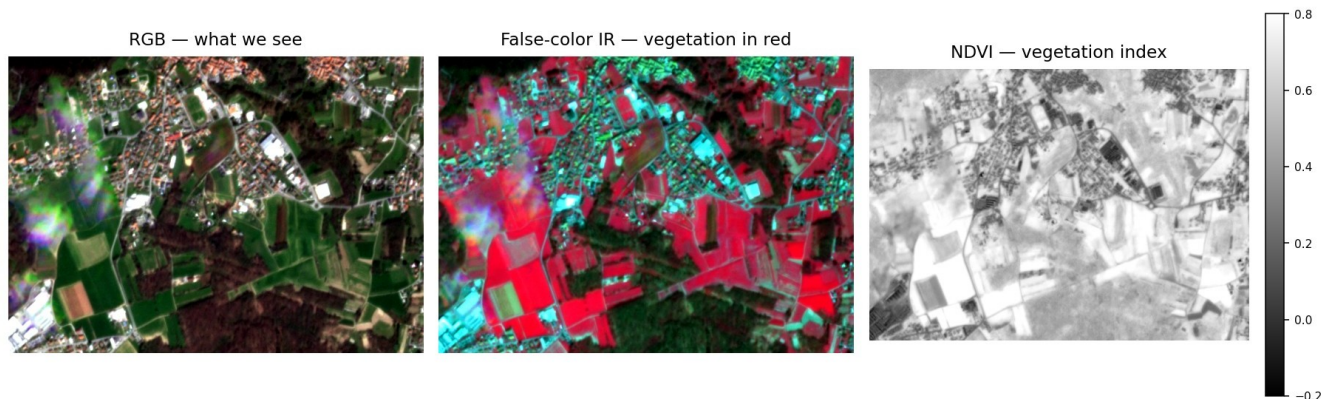


From bands to information: three views of the same scene

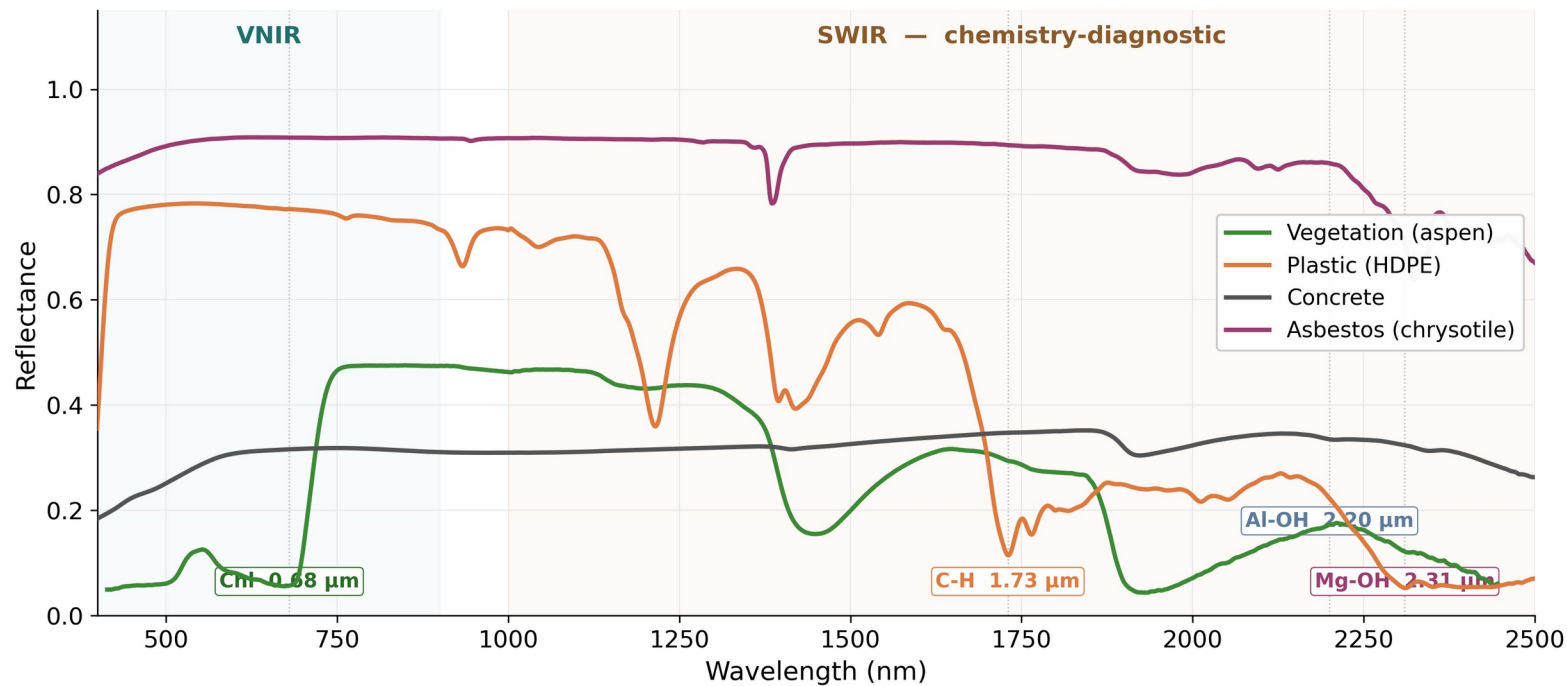
Same multiband image, many ways to read it.

- RGB, what we see.
- False-colour IR, vegetation in red.
- NDVI, vegetation index map.

Material discrimination needs more than RGB.



Every material has a spectral fingerprint

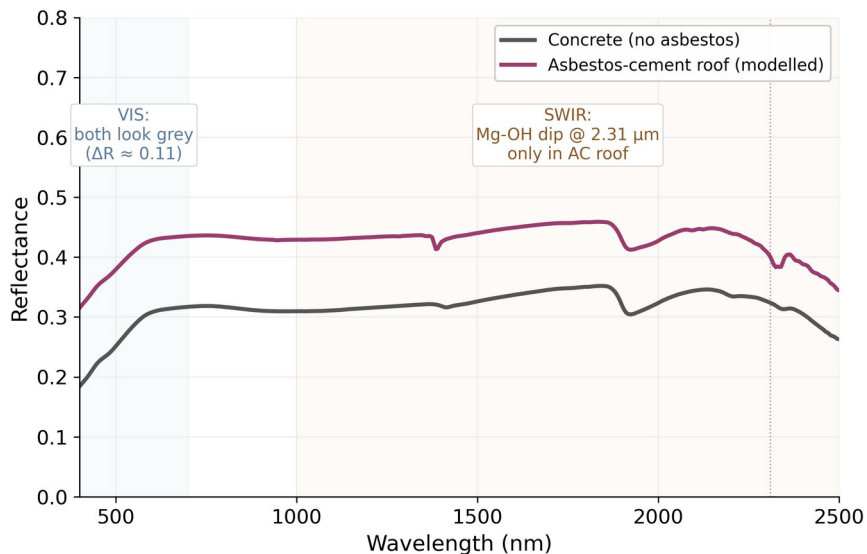


USGS splib07a (Kokaly et al., 2017)

RGB fails in two distinct ways

iso-chromaticity and sub-pixel mixing, both invisible without spectral information.

Same color, different materials



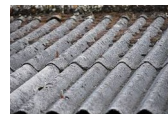
USGS splib07a (Kokaly et al., 2017)



HDPE plastic film



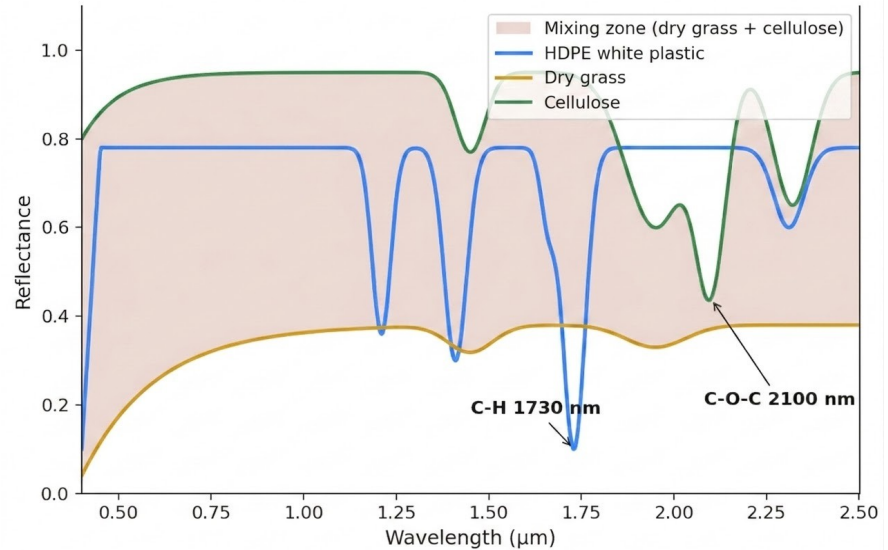
Asbestos-cement roof



Light concrete slab

Different materials mix into the target's magnitude

- A pixel at 10–30 m rarely contains one pure material.
- A linear mix of two endmembers can land on the spectral magnitude of a third, pure material.
- The mixture cannot be unmixed in RGB, only SWIR features (C-H, C-O-C) keep the materials separable.



State of the art: asbestos discrimination

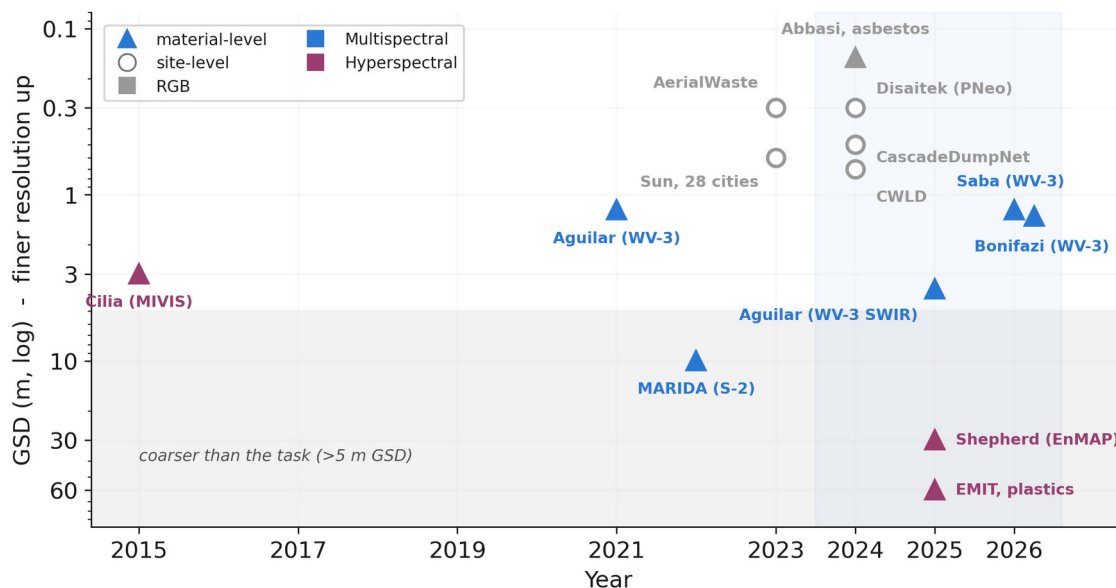
Work (year)	Input / GSD	Task	Method	Key result
Saba 2026 †	WV-3 VNIR 1.24 m	pixel classification	32 classifiers, Fine-KNN	Macro-F1 97.6% (VNIR only)
Bonifazi 2026 †	WV-3 16 b 1.24/3.7 m	pixel cls. + aggregation	Max-Likelihood, multi-temporal	AC roofs; removal monitoring
Shepherd 2025	EnMAP HSI 30 m	target detection	8-classifier cascade (ACE)	86% field match; OA 91.4%
Abbasi 2024 †	aerial RGB (no SWIR)	OBIA classification	DenseNet + LSTM, temporal	OA ~96%, AC 94% by shape+time
Cilia 2015 (hist.)	MIVIS airborne HSI 3 m	pixel classification	SAM + MNF, ISD index	PA 89% / UA 86%

State of the art: plastics & urban materials

Work (year)	Input / GSD	Task	Method	Key result
Aguilar 2025	WV-3 SWIR 3.7 m	target detection	matched filter (USGS spectra)	precision 92.5%; lab-image $r=0.95$
CDW (Vitek) 2025	lab HSI 768 b	classification	narrowband selection + MLP	RGB 0.87 $\rightarrow$ +2 NIR 0.96 ~ full HSI
EMIT (Estrela 2025)	EMIT HSI 60 m	target detection	matched filter, HDPE/PVC	first global orbital plastic map
SpectralWaste 2024	RGB + SWIR HSI (conveyor)	segmentation	RGB/HSI/fusion (CMX)	fusion mIoU 58 > RGB 48
MARIDA 2022	Sentinel-2 10 m	segmentation	RF / U-Net benchmark	F1 0.79; sub-pixel mixing at 10 m
Aguilar 2021 (hist.)	WV-3 VNIR + SWIR	OBIA classification	DT, 3-way band ablation	OA 90.85 $\rightarrow$ 96.79 $\rightarrow$ 97.38

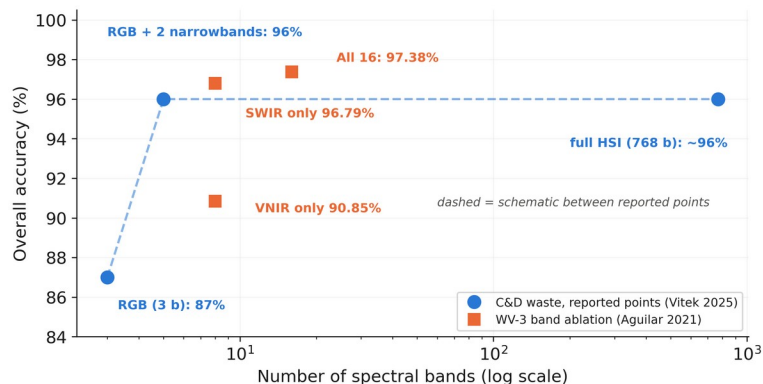
A moving field: material-level evidence is 2024-2026

Site-level RGB detection matured in 2023-2024; material-level discrimination is being demonstrated NOW (2024-2026) - but not yet inside an illegal-waste pipeline at task resolution.



More bands $\neq$ more information, well-chosen few suffice

- Vitek 2025 (lab HSI, C&D waste): RGB 0.87 $\rightarrow$ RGB + 2 NIR bands 0.96 - on par with the full 768-band HSI.
- SpectralWaste 2024: RGB+SWIR fusion beats either alone (mIoU 58 vs 48); SWIR wins the thin, iso-colour classes.
- Historical anchor - Aguilar 2021 (WV-3): VNIR 90.85 $\rightarrow$ All 97.38, and the jump came from SWIR.



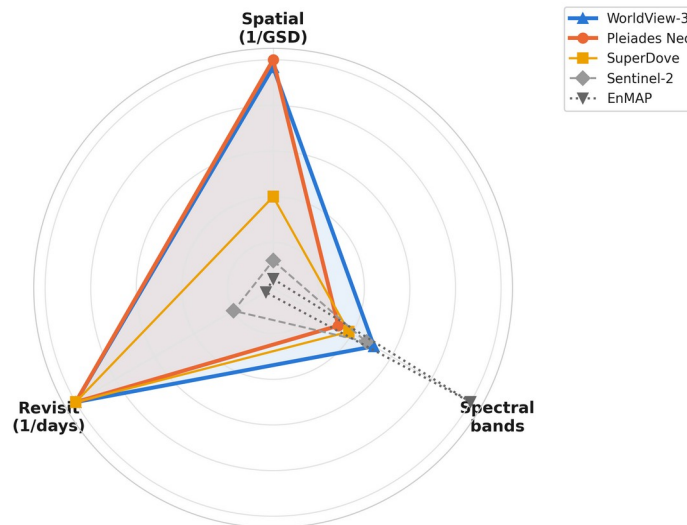
Why it matters: WHICH bands is material-dependent - VNIR suffices on some materials (Saba 2026); chemistry-bound hazards need SWIR.

Exactly what a controlled ablation must measure.

The sensor trade-off: spatial x spectral x revisit

Three axes, one photon budget. No satellite maximises all three.

- WorldView-3: 1.24 m VNIR + 3.7 m SWIR, 16 bands - the only VHR + SWIR option.
- Pleiades Neo: 1.2 m, 6 VNIR, no SWIR - sharpest VNIR-only.
- Sentinel-2 / EnMAP: 10-30 m - rich spectra, too coarse for material at site scale.
- SuperDove: 3 m, 8 VNIR, free, near-daily - wide-area but VNIR-only.



Outer ring = better. Spatial 1/GSD - bands log10(N) - revisit 1/days (vendor capability: WV-3 <1 d, PNeo daily, SuperDove daily).

Very-high resolution: the SWIR divide

	WorldView-3	Pleiades Neo	SuperDove
VNIR	8 b @ 1.24 m	6 b @ 1.2 m	8 b @ 3 m
SWIR	8 b @ 3.7 m	none	none
Pan	~0.31 m	~0.30 m	-
Access	commercial / TPM	commercial / TPM	free, near-daily

WorldView-3 is the only sub-2 m platform that also carries SWIR - where the chemistry lives.

Pleiades Neo and SuperDove are VNIR-only: sharp, but blind to the diagnostic SWIR absorptions.

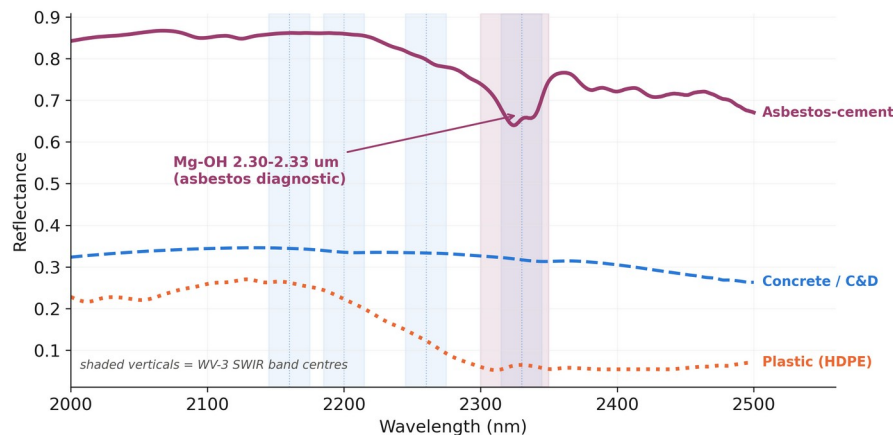
Precision (the Sentinel-2 trap): S-2 DOES carry SWIR, at 20 m - its limit here is resolution, not the band. The VHR question is who carries SWIR at all.

Where each hazard becomes separable: feature → band

	Pleiades Neo (VNIR-only)			WorldView-3 adds SWIR			
Asbestos (Mg-OH)						weak	strong
Plastics (C-H)				strong	strong	weak	weak
Concrete/C&D (carbonate)						strong	strong
Slag/rust (Fe-oxide)		weak	strong				
AC weathering (moss/lichen)		strong	weak				
	RGB	Red Edge	NIR	SWIR 1.2um	SWIR 1.73um	SWIR 2.2um	SWIR 2.33um

Honest limit: resolution is split between texture and chemistry

- Spectra are sampled coarsely: VNIR ~1.24 m, SWIR ~3.7 m (~14 m² pixels).
- The pan band gives fine texture ~0.3 m, but no material info alone.
- A small dump is easier to SEE (texture) than to chemically IDENTIFY (spectra).
- SWIR-8 bottleneck: asbestos 2.32 + concrete 2.34 + plastic 2.31 crowd one band.
- And SWIR is not always needed: Saba 2026 / Abbasi 2024 classify asbestos at VHR with none.



State of the art: datasets & the data gap

Work (year)	Input / GSD	Task	Method	Key result
CWLD 2024	GF-2 0.8 m + GE 0.5 m	segmentation	C&D masks, DeepLabV3+	F1 88.9% / IoU 82% (China)
SpectralWaste 2024	RGB + SWIR HSI	segmentation	conveyor-belt, 224-band	sorting plant, not Earth observation
AerialWaste (Torres 2023)	aerial RGB 20-50 cm	classification	22 material categories	RGB-only; coordinates withheld
MARIDA 2022	Sentinel-2 10 m	segmentation	15-class pixel benchmark	marine debris - water, not land

Foundation models for Earth Observation (EO): a new lever

2024-2025: a wave of pretrained backbones for EO (Prithvi, SpectralGPT, DOFA, AnySat).

Most are pretrained at 10-30 m on Sentinel-2 / HLS - naive transfer to sub-metre VHR is not guaranteed.

Two routes forward

- Sensor-agnostic models (DOFA, AnySat), accept arbitrary band sets.
- Parameter-efficient adapters (DEFLECT), freeze the backbone, add small heads (<1% parameters).

Open question: does pretraining at 10-30 m transfer to ~1 m VHR - and to material classification?

State of the art: foundation models

Work (year)	Input / GSD	Task	Method	Key result
AnySat (Astruc 2025)	11 sensors 0.2-250 m	pretraining	JEPA multimodal	SOTA on 9 tasks
DEFLECT 2025	MS 6-13 b	adaptation	PEFT adapter <1% params	on par with full fine-tune
DOFA (Xiong 2024)	5 modalities 1-30 m	pretraining	wavelength hypernetwork	202 bands; new-sensor plug-in
SoftCon 2024	Sentinel-1/2 13 b	pretraining	soft contrastive + DINOv2	SOTA 10/11; random-init extra bands
Prithvi-EO-2.0 2024	HLS 6 b incl. SWIR 30 m	pretraining	3D temporal MAE	+8 pp GEO-Bench; fixed bands

DOFA: a band-agnostic backbone, the candidate to start from

Dynamic One-For-All (Xiong et al. 2024).

- A hypernetwork generates patch-embedding weights from each band's central wavelength.
- One backbone ingests any sensor (WV-3 16 b, Pleiades Neo 6 b, Sentinel-2 13 b, HSI) by wavelength.
- Makes an RGB vs VNIR vs SWIR comparison fair - the substrate for a controlled band ablation.
- Pairable with PEFT adapters (DEFLECT, <1% params) for cheap spectral fine-tuning.

State of the art: object vs material

Work (year)	Input / GSD	Task	Method	Key result
ELV Hybrid-YOLOv5 2025	close-range infrared	object detection	Hybrid-YOLOv5	mAP 84%; needs spectrum / close-range
Ramachandran 2024	VHR sub-metre	object detection	DL detector (Nat. Comm.)	tanks P 0.96 / R 0.97; >169k mapped
YOLOv7-OT 2024	VHR satellite	object detection	YOLOv7 + CBAM	tanks 90% acc / 95.9% precision
UAV solid-waste 2024	UAV	segmentation	dual-branch network	OA >94%; generic pile, no per-material

Gaps the state of the art reveals

- RGB ceiling: every operational VHR waste detector is RGB-only and learns shape/context, not material.
- No public dataset combines VHR + terrestrial waste + material labels (AerialWaste is RGB; CWLD is C&D-only).
- Material evidence is adjacent, not direct: proven on greenhouses / asbestos roofs in isolation, never in a waste pipeline.
- Generalization (cross-region / cross-sensor / cross-time) is fragile and unmeasured for MS.
- Foundation models pretrain at 10-30 m and are untested for material at VHR.
- No link from material → hazard class → risk, the decision output agencies actually need.

Generalization: fragile and largely unmeasured for MS

	In-domain	Cross-region	Cross-sensor
RGB (today)	strong	-5.1% F1 (Gibellini)	ports trivially
+ VNIR	-	gap narrows (physics)	band-shift fragile
+ SWIR	-	gap narrows most	widens if not harmonized

A recurring SOTA blind spot: most works report in-domain only. Whether added bands narrow or widen the gap is itself an open question.

What the state of the art points to

The open question is sharp and unanswered: how much does multiband beat RGB for waste MATERIAL - and where does it stop?

- A controlled band ablation (RGB → +RedEdge/NIR → full VNIR → +SWIR) on a wavelength-agnostic backbone (DOFA).
- Generalization as a first-class axis (cross-region / sensor / time), not an afterthought.
- Candidate VHR platforms span the SWIR divide: WorldView-3 (with SWIR) vs Pleiades Neo (VNIR-only).
- Asbestos is the cleanest first material: public ground truth + a textbook SWIR signature.

Independent of any single dataset: the SOTA defines the experiment, not the data in hand.